

CHAINING

CHARACTERISTICS OVERVIEW CHART

Verbal Skills	Grade Levels	Cognitive Level	Areas Addressed
☒ Nonverbal	☒ PK	☒ Classic	☐ (Pre)Academic/Cognitive/Academic
☒ Mixed	☒ Elementary	☐ High Functioning	☒ Adaptive Behavior/ Daily Living
☒ Verbal	☒ Middle/High		☒ Behavior ☐ Communication/Speech ☒ Social/Emotional

BRIEF INTRODUCTION

Chaining is an instructional strategy grounded in applied behavior analysis (ABA) theory.

Chaining is based on task analysis, whereby sub-behaviors are recognized as requirements for task mastery.

DESCRIPTION

Chaining breaks a task down into small steps and then teaches each step within the sequence by itself. For example, a child learning to wash her hands independently may start with learning to turn on the faucet. Once this skill is learned, the next step may be rinsing hands, etc. This technique is helpful in assisting children to learn a routine task that is repetitive, such as using the bathroom, brushing teeth, putting on clothes and shoes, or completing a work task.

In utilizing chaining, the teacher must (a) analyze the desired behavior, (b) break it into steps, and (c) plan for the teaching. There are two type of chaining techniques:

1. *Forward chaining.* The forward chaining technique moves a child from the first part of the task to the end. In short, each step must be mastered before the next step in the skill series is added.
2. *Backward chaining.* The backward chaining technique involves the same process as forward chaining, except in reverse. That is, the teaching process moves from the last

part of the task to the beginning. This technique is used when it is easier to teach a child a task from the last step than from the beginning.

STEPS

The steps in forward chaining:

1. The teacher teaches the child the first step in the chain.
2. When the first step is learned, the teacher adds the second step. The child is learning the second step in the routine and attaching it to the first step.
3. The third step is taught in conjunction with the first two steps once the child is able to demonstrate the first two steps.

The steps in backward chaining:

The steps in backward chaining mirror those of forward chaining except they occur in reverse.

The teacher provides the child assistance throughout the process until the last step.

1. The child is encouraged to complete the last step independently.
2. When the last step is mastered, the teacher provides assistance until the child is able to perform the step before the last one.
3. The child completes more and more ending steps independently until he masters all steps and is able to complete the task without assistance.

BRIEF EXAMPLE

One of 6-year-old Tommy's goals was to learn to brush his teeth. In order to teach Tommy this skill, his teacher decided to use backward chaining. First, the teacher conducted a task analysis to identify all the steps of tooth-brushing. In the beginning, the teacher guided and prompted Tommy to perform all steps except the last one: looking at the mirror and wiping his mouth with a towel. Tommy learned to perform this last step first with prompts and reinforcement. After mastering this last step, the next teaching target was the second-to-the-last step, turning off the

faucet. With backward chaining, Tommy learned to brush his teeth by working on the last step first.

SUMMARY

Chaining is a behavioral strategy used to teach children with autism complex behaviors by breaking them down into smaller sequential steps. One of two methods, forward chaining and backward chaining, is selected based on the nature of the task or the skill levels of the child.

RESEARCH TABLE

Number of Studies	Ages (year)	Sample Size	Area(s) Addressed	Outcome
6	3 to adult	20	Daily-living skills (drinking from a cup, Internet skills, making a snack, independent eating) escape behavior, sharing	+

STUDIES CITED IN RESEARCH TABLE

1. DeQuinzio, J. A., Townsend, D. B., & Poulson, C. L. (2008). The effects of forward chaining and contingent social interaction on the acquisition of complex sharing responses by children with autism. *Research in Autism Spectrum Disorders, 2*, 264-275.
Four children with autism (8 to 10 years) were taught a sharing response using forward chaining. None of the participants engaged in the sharing response chain during baseline. Systematic increases occurred for all four participants during the intervention. In addition, the children generalized responses across settings, peers, and toys.
2. Jerome, J., Frantino, E. P., & Sturmey, P. (2007). The effects of errorless learning and backward chaining on the acquisition of Internet skills in adults with developmental disabilities. *Journal of Applied Behavior Analysis, 40*, 185-189.

Three adults with autism and mental retardation were taught to access specific Internet sites using backward chaining and most-to-least intrusive prompting. Results indicated that the number of independent steps completed in the task analysis increased following training.

3. Hagopian, L., Farrell, D. A., & Amari, A. (1996). Treating total liquid refusal with backward chaining and fading. *Journal of Applied Behavior Analysis, 29*, 573-575.

This study reported on a 12-year-old boy with autism, mental retardation, and a history of severe gastrointestinal problems who presented with total refusal of liquids and food. Backward chaining was used to shape drinking from a cup, and a fading procedure was used to increase the quantity of water he was required to drink. The study illustrated how a chain of responses that is totally absent can be shaped by first targeting a simple preexisting response in the chain.

4. Lalli, J. S., Casey, S., & Kates, K. (1995). Reducing escape behavior and increasing task completion with functional communication training, extinction, and response chaining. *Journal of Applied Behavior Analysis, 28*(3), 261-268.

The effects of functional communication training, extinction, and response chaining on participants' (10, 13, 15 years of age) escape behavior by evaluating the effects of response chaining following the implementation of FCT. Participants were taught an escape response and then a response chaining procedure to increase their participation in the task. Results showed that the treatment reduced rates of aberrant behavior and that the chaining procedure was effective in decreasing the availability of escape.

5. Kayser, J. E., Billingsley, F. F., & Neel, R. S. (1986). A comparison of in-context and traditional instructional approaches: Total task, single trial versus backward chaining, multiple trials. *Journal of the Association for Persons with Severe Handicaps, 11*, 28-38.

This study compared the effects of total-task/single-trial and backward-chaining/multiple-trials instruction to teach eight children with severe disabilities how to make a snack. Total-task/single-trial instruction resulted in superior acquisition of independent steps in the training setting for three of the eight children and less substantial effects for two participants. For three participants, differences between the two methods were negligible or nonexistent in terms of independent steps performed. In all cases, instructional time was substantially less for total-task/single-trial instruction than for backward-chaining/multiple-trials instruction. Findings related to generalization were mixed.

6. MacArthur, J., Ballard, K. D., & Artinian, M. (1985). Teaching independent eating to a developmentally handicapped child showing chronic food refusal and disruption at mealtimes. *Australia & New Zealand Journal of Developmental Disabilities, 12*, 203-210.

This study described an intervention to establish independent eating behaviors in a 3-year-old boy with autistic-like behaviors. The first phase of the intervention, using backward chaining with prompting and fading of prompts, reduced screaming, food refusal, and related disruptive mealtime behaviors and established appropriate eating responses. The second phase of the intervention successfully taught the mother to implement the teaching strategies, and independent mealtime behaviors were established at home. In addition, follow-up showed maintenance of independent eating.

REFERENCES

- DeQuinzio, J. A., Townsend, D. B., & Poulson, C. L. (2008). The effects of forward chaining and contingent social interaction on the acquisition of complex sharing responses by children with autism. *Research in Autism Spectrum Disorders, 2*, 264-275.
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RESOURCES AND MATERIALS

- Interactive Collaborative Autism Network - www.autismnetwork.org/modules/behavior/chaining/index.html
This link takes the user to an interactive educational module that includes a pre- and posttest as well as content that includes the steps to forward and backward chaining.
- Chaining – <http://www.polyxo.com/discretetrial/complex.html>.
This web site gives a brief description and example of chaining.

GENERAL RESOURCES

- Autism Internet Modules (AIM) www.autisminternetmodules.org
The Autism Internet Modules were developed with one aim in mind: to make comprehensive, up-to-date, and usable information on autism accessible and applicable to educators, other professionals, and families who support individuals with autism spectrum

disorders (ASD). Written by experts from across the U.S., all online modules are free, and are designed to promote understanding of, respect for, and equality of persons with ASD.

Current modules are:

- Assessment for Identification
 - Home Base
 - Peer-Mediated Instruction and Intervention (PMII)
 - Picture Exchange Communication System (PECS)
 - Pivotal Response Training (PRT)
 - Preparing Individuals for Employment
 - Reinforcement
 - Restricted Patterns of Behavior, Interests, and Activities
 - Self-Management
 - Social Supports for Transition-Aged Individuals
 - Structured Teaching
 - Structured Work Systems and Activity Organization
 - Supporting Successful Completion of Homework
 - The Incredible 5-Point Scale
 - Time Delay
 - Transitioning Between Activities
 - Visual Supports
- Interactive Collaborative Autism Network (iCAN) <http://www.autismnetwork.org>
iCAN offers free online instructional modules on autism spectrum disorder (ASD). Modules have been developed in these areas:
 - Characteristics
 - Assessment
 - Academic Interventions
 - Behavioral Interventions
 - Communication Interventions
 - Environmental Interventions
 - Social Interventions
- Indiana Resource Center for Autism (IRCA) <http://www.iidc.indiana.edu/irca/fmain1.html>
The Indiana Resource Center for Autism staff 's efforts are focused on providing communities, organizations, agencies, and families with the knowledge and skills to support children and adults in typical early intervention, school, community, work, and home settings.
 - IRCA Articles <http://www.iidc.indiana.edu/irca/ftrainpapers.html>
 - IRCA Modules <http://www.iidc.indiana.edu/irca/fmodules.html>
- Texas Statewide Leadership for Autism www.txautism.net
The Texas Statewide Leadership for Autism in conjunction with the network of Texas Education Service center with a grant from the Texas Education Agency has developed a series of free online courses in autism. Please check the training

page, www.txautism.net/training.html, for update lists of courses, course numbers and registration information. Current courses include the following:

- Autism 101: Top Ten Pieces to the Puzzle
- Autismo 101: Las 10 piezas principales del rompecabezas
- Asperger Syndrome 101 Online
- Asperger Syndrome 101 Online
- Navigating the Social Maze: Supports & Interventions for Individuals with Autism Spectrum Disorders
- Communication: The Power of Communication for Individuals with Autism Spectrum Disorders
- Communication: The Power of Communication for Individuals with Autism Spectrum Disorders

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Improving Behavior through Differential Reinforcement: A Praise Note System for Elementary School Students

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Abstract

Schools are often in need of low-cost, high-impact strategies to improve student behavior in school common areas. While many behavior management programs exist, there are few resources available to guide the implementation of these programs and ensure they are grounded in evidence-based strategies. Therefore, the current study had two primary purposes: first, to demonstrate the effectiveness of a simple behavior management system, and second, to begin the process of providing some guidance for the application of similar systems. The study used a differential reinforcement of incompatible behavior (DRI) procedure in a multiple-baseline design across three target behaviors to decrease inappropriate and increase appropriate behaviors in an elementary school lunchroom. The intervention consisted of 1) teaching specific appropriate behaviors 2) providing opportunities for students to practice skills, and 3) implementing a Praise Note system to reward students for behaving appropriately. Students were taught appropriate behaviors, and the school staff was trained to recognize and reward students who kept the lunchroom clean, sat appropriately in their seats, and walked in the lunchroom. Data show significant decreases for each of three target behaviors. The average amount of litter left in the lunchroom decreased by 96%, the average number of instances of sitting inappropriately decreased by 64%, and the average number of instances of running in the lunchroom decreased by 75%. Beyond contributing to the effective intervention strategies available to reduce problem behaviors in common areas, this research also provide much needed guidance for effectively implementing the necessary components of a Praise Note system, namely, program intensity, and delivery.

Behavior problems are a well-documented and an ever-increasing challenge facing educators (Mayer, 2001; Walker & Horner, 1996) and effective behavior management strategies that meet these challenges and promote safety for students and adults are of

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significant interest for educators nationally (Scott, 2001; Turnbull, Edmonson, Griggs, Wickham, Sailor, Freeman, Guess, Lassen, McCart, Park, Riffel, Turnbull, & Warren, 2002). To confront these problems, many administrators have already successfully adopted school-wide intervention strategies to address concerns throughout the school environment (Lewis & Sugai, 1999; Oswald, Safran, & Johanson, 2005).

Most traditional behavior management programs have focused on teaching and rewarding appropriate student behaviors that typically occur in the classroom setting. While these strategies may be effective within the classroom, they often fail to address concerns associated with misbehavior in school common areas. School "common areas," can be described as places where relatively few adults supervise large groups of students, and little or no formalized instruction occurs. School common areas may include hallways, lunchrooms, playgrounds, and buses or bus lines.

Previous research suggests that misbehavior in school common areas accounts for approximately one-half of all problem behaviors in many schools (Colvin, Sugai, Good, & Lee, 1997). In an assessment of office discipline referral (ODR) frequency by location, 47% of all office-managed discipline referrals collected throughout the school year resulted from problems arising in common areas (Todd, A., Haugen, L., Anderson, K., & Spriggs, M., 2002). The average elementary school student with a thirty-minute lunch break, two ten-minute recesses, and occasional trips to and from the library or other areas of the school, may spend up to one hour or more of a typical school day in common areas (this number is increased for students who ride the school bus or participate in resource or special education programs). Therefore, given the amount of time students spend in common areas and the number of overall problem behaviors that reportedly occur there, it is estimated that problem behavior is six times more likely to occur in common areas than in the classroom.

Various circumstances have been identified as possible contributors to the high number of problem behaviors evidenced in school lunchrooms and other common areas. Some of the most significant factors include large numbers of students, a large amount of physical space to monitor, and too few adults trained to effectively deal with problem behaviors (Todd et al., 2002). The transition from structured classroom environments to unstructured common areas and back again several times throughout the school day may also pose a difficult task for many students. High frequency transitions and shifting expectations across school common areas can lead to confusion that can inflate the number of problem behaviors.

Lunchrooms, one of the most significant school common areas in regard to capacity and time of use, may be particularly subject to student misbehavior. Therefore, lunchroom group management issues present a valid concern for many teachers, school administrators, and even students. Historically, inappropriate lunchroom behaviors have included talking while the aide is talking, being out-of-seat, and quarreling (MacPherson, Candee, & Hobman, 1974). Recent research confirms that concerns in today's lunchrooms are relatively similar to those in the past. In a survey conducted as part of a study by Samuels, Swerdlik, and Margolis (2001) common student complaints of the lunchroom consisted of excessive noise level, students running and shouting, and lunchroom supervisors yelling at students. Teacher concerns included confusion, messiness, and high noise level. While dealing with these behaviors in the lunchroom is of primary concern, evidence also suggests that efforts to improve lunchroom behavior may lead to improvement in other aspects of the learning environment. In the previously mentioned study conducted by Samuels et al. (2001), a decrease in the number of instances of inappropriate and disruptive behaviors in the lunchroom was accompanied by a decrease in the number of office discipline referrals in periods before and after lunch.

As a means of coping with the current circumstances, educators are in need of high-impact strategies for improving student behavior and contributing to school environments that are more conducive to learning. Programs using positive methods of supporting appropriate behavior are highly effective in helping schools accomplish this goal. A large body of research evidence has documented the effectiveness of these programs in a wide variety of school circumstances for both individual students (Carr, Horner, Turnbull, Marquis, Magito McLaughlin, McAtee, Smith, Anderson Ryan, Ruef, & Doolabh, 1999; Todd, Campbell, Meyer, & Horner, 2008) and groups of students (Lewis, Sugai & Colvin, 1998; Sugai & Horner, 1999) in common areas (Kartub, Taylor-Greene, March, & Horner, 2000; Lewis, Colvin & Sugai, 2000; Nelson, Colvin & Smith, 1996; Todd et al., 2002) as well as in the classroom (Skinner, Neddienriep, Robinson, Ervin, & Jones, 2002; Todd et al., 2008). These programs are consistent with research findings that suggest the most effective programs for promoting lasting change in student behavior focus on the development of interventions that will be implemented consistently across all school settings (Lewis & Sugai, 1999; Walker & Horner, 1996; West, 1993; West, Young, Mitchem, & Caldarella, 1998). One framework that is currently gaining momentum with schools all over the country is Positive Behavior Support (PBS) as outlined by Carr, Dunlap, Horner, Koegel,

Turnbull, Sailor, Anderson, Albin, Koegel, & Fox (2002). PBS provides schools with a philosophical approach for developing and implementing programs that use positive methods to improve student behavior. Because PBS is an approach rather than a set of specific programs and practices, there is considerable variability in the types of programs implemented at the school level. Therefore, programs implemented at many schools may fit the PBS theme but may not necessarily be evidence-based. To improve the quality and effectiveness of these programs, researchers should strive to provide teachers and administrators with evidence-based guidelines for implementing PBS programs. For example, program intensity and coverage standards are needed, training procedures should be well documented, and data collection strategies should have planned interobserver reliability checks.

There are several evidence-based strategies currently available to educators that fit the PBS model. Some methods relevant to the current study will be discussed in more detail. They are: providing clear expectations and using common language, consistently delivering praise (written and verbal), encouraging active supervision, implementing token economies, and utilizing DRI—the differential reinforcement of incompatible behaviors (West & Sloane, 1986).

Providing clear expectations for students is an essential part of a behavior support program. Providing clear expectations may consist of defining rules and ensuring that students and teachers use common language to describe behaviors and consequences. In one study targeting common areas, Kartub et al. (2000) outlined an intervention to reduce hallway noise during transition times in a rural middle school by clarifying student expectations. This intervention was developed to help students discriminate a difference in hallway expectations before and after school (when students were allowed to be noisy) and during transition times (when students were expected to maintain quieter noise levels). As part of the intervention, the hallways were made to look different during transition times with the addition of small blinking lights. Noise levels in the hallway were measured during three phases: baseline, a noise reduction phase, and a follow-up phase. During baseline the average decibel levels were 74.8, 76.5, and 76.8 for the sixth, seventh, and eighth graders, respectively. During the noise reduction phase the average decibel levels fell to 67.4, 68.6, and 68.9. The overall noise level decreased and stayed within an acceptable range after the intervention was employed.

In another example, a school-wide intervention effectively decreased problem behaviors in the hallway including running, walking out of line, invading another's space, walking on the wrong side of the hallways, and yelling (Leedy, Bates, & Safran, 2004). In this study,

teachers and university faculty members developed clear behavioral expectations and promoted these expectations throughout the school. With the combined support of schoolteachers and administrators, the intervention led to a combined increase of 134.9% in appropriate student behavior.

The delivery of contingent and specific praise is another evidenced-based strategy for improving student behavior. In a study conducted by Madsen, Becker, and Thomas (1968) teacher behavior was modified to show the individual effects of classroom rules, ignoring, and praise on student problem behavior. In the first two phases of the study teachers were taught to remind students of rules and to ignore problem behavior. In the third phase teachers were taught to deliver general and specific praise to students for rule following and other prosocial behaviors. Results from this research clearly showed that introduction of rules alone had little effect on problem behavior, ignoring inappropriate behavior produced inconsistent results, and ignoring accompanied by praise was very effective. The authors concluded that praise was the "key teacher behavior in achieving effective classroom management."

Active supervision has been used effectively to reduce problem behaviors in hallways, lunchrooms, and on school playgrounds at both primary and secondary schools (Colvin et al., 1997; Johnson, Lyons, & Griffin, 2008; Lewis, Colvin, & Sugai, 2000). Colvin, Sugai, Good and Lee (1997) define active supervision as "specific and overt behaviors (scanning, escorting, interacting) displayed by supervisors designed to prevent problem behavior and promote rule following behavior." Some elements of active supervision may include: reinforcing rule compliance, error correction for rule violation, and physical movement around and scanning of the area being supervised (Lewis, Colvin & Sugai, 2000). Colvin, Sugai, Good, and Lee (1997) used active supervision and precorrection to improve student behavior in common areas in a rural elementary school containing 475 students. In this demonstration, supervisory staff members were trained to increase active supervision strategies. Staff training included explanation, demonstration, and role-playing of active supervision and precorrection strategies. In addition, staff members received twice-monthly reminders concerning use of active supervision procedures throughout the course of the study. The study used a multiple baseline design across three transition areas to measure the effects of the intervention on the instances of problem behavior. The data demonstrate a decrease in the number of problem behaviors exhibited by students across all transition areas. In addition, a significant correlation ($-.83$) was found between the number of student-teacher interactions and frequency of

problem behavior. In other words, the greater the number of interactions that staff had with students, the fewer problem behaviors were exhibited.

The use of token reinforcers is another effective strategy for managing student behavior. Token reinforcers are "symbolic representations exchangeable for some reinforcer of value to students" (Alberto & Troutman, 2003). Tokens function in a manner very similar to the way that money functions in general society; they have little or no inherent value and are usually delivered immediately following the occurrence of behavior. At a convenient time tokens are exchanged for desired reinforcers. In this way they serve as a transition between the behavior and reinforcement. Token systems in many forms are pervasive in the educational system today, and are used in many resource and self-contained classrooms, as well as in many general education classrooms (Alberto & Troutman, 2003). In one example of a token reinforcement program O'Leary, Becker, Evans and Saudargas (1969) looked at the various effects of several different intervention strategies on problem behavior. During the token reinforcement phase of the study students were allowed to earn points or tokens at designated times during the school day based on their behavior during a given time period. The total number of points each student earned ranged from one to ten and was based on the extent to which he or she followed the rules that were written on the blackboard. The points were later exchanged for backup reinforcers such as candy, comics, and small toys. Results of this research showed no consistent effects on behavior in any phase other than the token economy. A statistical analysis of the group data showed that the token reinforcement was associated with a significant reduction in the number of problem behaviors during that phase.

Another very effective strategy is the differential reinforcement of incompatible behavior (DRI). In this procedure a target behavior that the teacher wants to reduce is defined. Then a behavior is chosen for increase that is mutually exclusive with the target behavior to be decreased. For example, if out-of-seat behavior is the behavior to be decreased, the behavior to be increased and reinforced is in-seat behavior. As the student is consistently reinforced for appropriate behavior, it increases and, as a result, the problem behavior decreases (Alberto & Troutman, 2003). The DRI procedure, in conjunction with increased response cost, was used to increase the attentive behavior of 30 third-grade and fourth-grade students at an elementary school (Zaghalwan, Ostrosky & Al-Khateeb, 2007). The sample used in the previously cited study, included 60 students from eight different elementary schools who scored highest on an ADHD checklist completed by

their resource room teachers. The researchers randomly assigned the students to either a treatment or control group. Students in the treatment group earned smiley faces for displaying attentive behaviors during instructional sessions. If students displayed inattentive behaviors smiley faces were removed and reinforcers were harder to earn. Analysis of Variance (ANOVA) results revealed an effect $\{F(1,57) = 165.20, p < .0001\}$ in favor of the experimental group, indicating that the intervention was successful in decreasing inattentive behaviors.

The previous examples of behavior management techniques represent a small sample of the evidence-based strategies that are available to educators in general and special education classrooms. While there is an abundance of literature outlining specific programs and case studies utilizing these various strategies, there is often little or no guidance available regarding best practice in their application.

One example of this problem is illustrated in the widely-used system of a "good ticket" program currently used in many schools. The concept of ticket systems is not new, and many schools employ similar programs with varying levels of success. For example, these systems are described in terms of their application as "Gotcha" coupons, "Chance" tickets, "High Fives," or "Pride tickets" (Oswald et al, 2005). In most instances these "notes" or "tickets" function as token reinforcers that are often entered into a lottery or exchanged for desired rewards. While research utilizing similar approaches does exist, there is a strong need for studies that outline minimum basic standards for program duration, intensity, tracking, and reinforcer delivery.

The current research presents a Praise Note system similar to the ticket systems discussed previously, with a few variations. The goal of this study is to demonstrate the effectiveness of a simple behavior management system based on the strategies outlined earlier in this article and to provide practitioners with guidance for the implementation and evaluation of similar systems.

Method

Participants and Setting

Participants in this study were approximately 200 first through fifth grade students in a rural northern Utah elementary school. The students were predominantly white (80%), and from middle to lower-middle socioeconomic backgrounds.

The targeted setting for this study was the school lunchroom, and all data collection took place during the regularly scheduled lunch period. All first through fifth grade students ate lunch at approximately the same time each day. Three grades were typically in the lunchroom at the same time (approximately 20 minutes for each

grade), with younger grades rotating out of the lunchroom as the older grades came in. Thus, the first grade exited the lunchroom as the fourth grade entered and the second grade exited as the fifth grade entered.

The lunchroom was roughly divided into two sides. One side consisted of five rows of tables (with benches on each side) where students sat to eat their lunches (see Figure 1). The other side of the room contained the lunch serving area, trashcans, and tray disposal counter. Students entered the lunchroom, in lines, through one of two entrances located on the east wall, and proceeded to the serving area where they picked up a tray and received their lunch. Students continued in their lines to a designated table where they sat and ate. As the students finished they took their lunch trays to the disposal counter and exited through a door in the north wall.

Design

This study employed a multiple baseline design (Cooper, Heron, & Heward, 2006) across three targeted behaviors deemed inappropriate for the lunchroom by the school principal and faculty: littering, inappropriate sitting, and running. For the purpose of this study the definitions of these behaviors were as follows. *Littering* consisted of leaving on the floor (all litter left on the tables was swept onto the floor by lunchroom staff) any object that was larger (length or width) than a 3X5 inch index card. *Inappropriate sitting* included standing while eating or sitting with "back pockets off of the seat." *Running* was defined as having both feet leave the floor at the same time. These definitions and their rationale are further clarified in the measurement section of this article.

Procedure

The Praise Note system chosen for this intervention was part of a school-wide effort to decrease problem behaviors and increase incompatible appropriate behaviors in the lunchroom. The principal, teachers, and lunchroom staff identified three target behaviors critical to student safety and promoting a positive lunchroom environment. The current intervention consisted of two components: 1) providing students and faculty with clear expectations for lunchroom behavior, and 2) implementing a Praise Note system to reward student behaviors meeting the expectations.

Baseline. During the baseline condition no programs were in place at the school to improve student behavior in the lunchroom. Lunchroom staff members circulated through the lunchroom correcting students at their own discretion. As no training had been provided to

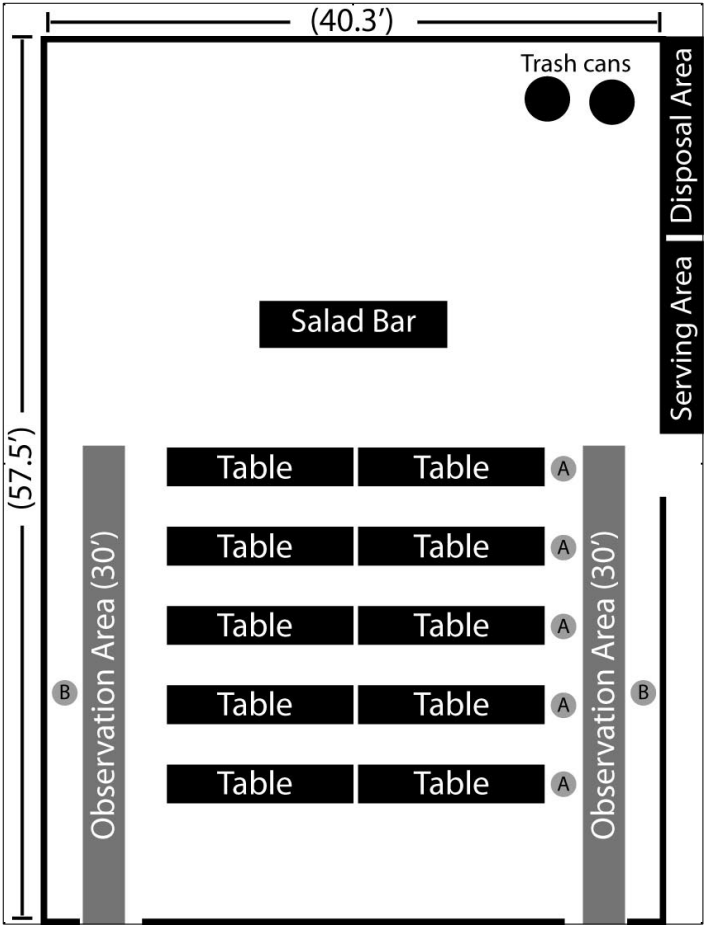


Figure 1. Diagram of the school lunchroom. The labels “A” and “B” represent the positioning of observers.

staff members concerning the delivery of praise statements, the rate of praise statements delivered to students during baseline was very low. In fact, the observers did not record any instances of praise statements happening during the baseline condition.

The Praise Note system. Students, teachers, custodians, lunchroom staff, and school administrators all participated in training sessions. After the sessions were completed, teachers and staff members were instructed to reward students who displayed appropriate behaviors in the lunchroom by presenting them with a Praise Note, accompanied by specific, verbal praise. Praise Notes were small yellow slips of paper that contained student and teacher signature lines, the school mascot (an eagle), and the school motto ("safe, kind, and responsible").

To ensure that Praise Notes were always available, faculty members received blank Praise Notes in their boxes, Praise Notes were also available in the office, and the principal or custodian always had some Praise Notes in the lunchroom. Faculty members and administrators in the lunchroom were encouraged to deliver one Praise Note every two to three minutes (VI 2.5 min).

When giving a Praise Note delivery the teacher filled in the student's name and signed the Praise Note. The student then took the Praise Note to the office during the course of that school day. Office staff members provided a large decorated jar for the students to place the Praise Notes in, and they congratulated each student who put a note in the jar. At the end of each school day the principal drew five Praise Notes from the jar and read the names of the students over the intercom system. Students, whose praise notes were selected, would then go to the office where they were allowed to choose a small reward (e.g., pencil, eraser, folder, sticker, etc.) from a box in the principal's office. To increase faculty participation, the teachers whose names were on the five selected Praise Notes were also entered into a weekly drawing for a gift certificate or service voucher donated by businesses in the community (e.g. one loaf of homemade bread, one free haircut, etc.).

After the drawing, all of the Praise Notes that were turned in were stapled to a large board in the main hallway near the school entrance. By filling the board students worked collectively to earn a larger group reward (e.g., extra recess time, a party, or early lunch). After the board was filled and the students received their reward, the Praise Notes were taken down and the students worked toward filling the board again. The posting and group reward allowed students to receive recognition even if their Praise Notes were not selected in the drawing.

Training

Teachers and other selected staff members (lunchroom personnel, school custodians, and lunchroom monitors) were given an overview of the Praise Note program before the intervention began. Teachers and lunchroom staff members were taught when, how, and for which behaviors a Praise Note should be delivered. Staff members were instructed to deliver Praise Notes during the lunch period to students who were meeting the expectations for the target behavior. When delivering a Praise Note staff members were taught to state the student's name (or give some other initial positive interaction), carefully describe the behaviors for which the Praise Note was given, write the student's name on the note, and present it to the student. Staff members were instructed to look for appropriate behaviors and deliver Praise Notes on a constant basis while in the lunchroom. They were also encouraged to target students who were in need of extra help.

Three training sessions were held (excluding the staff overview of the Praise Note system), with one session focusing on each of the three target behaviors: littering, inappropriate sitting, and running. Each training session was conducted for all students and participating faculty members the day prior to the beginning of each phase of the intervention. Students and their teachers participated in training sessions with the other classes in their grade level. Classes were brought into the lunchroom where they participated in a four-part training session lasting approximately 20 minutes, which consisted of an introduction and three learning stations (approximately 5 minutes each).

Introduction. To begin each training session the new lunchroom expectations were introduced to the group. In each of the training sessions participants were given explicit definitions of what behaviors were and were not appropriate in the lunchroom, including examples and non-examples. They were also introduced to the Praise Note system and told how they could earn Praise Notes. Students were informed that when they were behaving according to the expectations they would be eligible to receive a Praise Note from one of the school faculty or lunchroom staff members. Students were also told that they would not receive a Praise Note every time they were acting appropriately, but that if they were consistently meeting expectations their chances to earn a Praise Note would increase. Students were also informed that they should deliver the Praise Notes to the office to be entered into a daily drawing for reward. After the drawing, all of the Praise Notes would be posted on a board in the main hallway, and when the board was filled they would receive a group reward such as an ice cream party or extra recess.

After the introduction, the group was divided into three equal subgroups and sent to one of three different learning stations. The stations were designed to provide modeling, role-play and practice opportunities, a check for understanding, and opportunities for reinforcement. Groups cycled through the stations in approximately 5-minute intervals.

Station One. At station one, a trainer reviewed the information presented in the introduction meeting and clarified expectations by allowing students to ask questions. The trainer also used a game to ensure students understood the expectations. During the game the trainer tossed a beanbag to students in the group, and the student who caught the beanbag was given a hypothetical situation and asked to determine if the student in the situation was eligible to receive a Praise Note.

Station Two. At station two, students participated in an activity that allowed them to demonstrate the target behaviors in a fun and positive way. The content and format for this station differed with each training, depending on the target behavior.

To discourage running and to help students understand the difference between walking and other inappropriate behaviors, students participated in a walking relay. Students in the group were divided into two equal lines, and asked to walk a course from one end of the lunchroom to the other. One student from the front of each line began the relay. Observers watched students' feet during the relay, and if both feet left the ground at any time the student was asked to go back to the beginning and try again. When the students walked the designated course without running, they tagged the hand of the next student, and he or she would then walk the course. This process was repeated until each student had the chance to practice walking by participating in the relay.

To discourage littering and encourage throwing trash in the trash receptacles, students participated in another relay. In this relay the students were divided into two groups. The relay course was then modeled for the students by the trainer. The course consisted of picking up a lunch tray with a napkin and a few strips of paper on it, walking to the trash can without spilling the papers, circling the trash cans, dumping the contents of the trays in the trash, stacking the trays neatly on the disposal counter, and walking back to tag the hand of the next student in line. This process was repeated until each student had the chance to practice disposing of trash properly.

A game was also used to discourage inappropriate sitting and encourage sitting appropriately at the lunch tables. This game reinforced the expectation that once students placed their tray on the table

they were to remain seated with “pockets on the seat” until they were finished with their lunch and ready to exit the lunchroom. In this game students were asked to form a line similar to the one they would form upon entering the lunchroom on any given day. The trainer modeled the correct procedure while explaining the game to the students. Students were instructed to walk past the serving area, move to the tables, and sit down single file. When one side of the table was filled students were instructed to fill the benches on the other side of the table, just as would be expected during a regular lunch period. Students were then instructed to sit with their “pockets on the seat” until all students were seated. After all students were seated, they practiced exiting the area correctly. The student at the head of the line was instructed to stand, exit the table, and form a line. As soon as the first student stood, the second followed and so on until all students had completed the process. The trainer watched the clock and timed each round of the game while correcting any inappropriate sitting. The students played the game three times, attempting to beat their previous time.

Station Three. At station three students were provided with a chance to practice the appropriate behaviors and answer questions about when a Praise Note would be given. As each student successfully demonstrated the appropriate behavior, he or she received a Praise Note. Each student received at least one Praise Note during this exercise. Students were reminded that they would not receive a Praise Note every time they displayed appropriate behavior, but that consistent appropriate behavior would increase the likelihood that they would receive a Praise Note.

Measurement

All observations were conducted during the regularly scheduled lunch period. The lunch period began at 11:55 a.m., when the first grade entered the lunchroom, and ended at 12:45 p.m. The daily observation period lasted from 11:55 a.m. to 12:32 p.m. After this time students had left the lunchroom to go out to the playground. Because the lunch schedule rotated and the number of students in the room varied (i.e. there were fewer students in the lunchroom at the beginning and end of the lunch period than in the middle as the grades rotated out), the number of behaviors occurring throughout the observation period fluctuated. Therefore, frequency of behavior over the entire lunch period was used as the dependent measure.

Littering. Littering consisted of leaving on the floor any object that was larger (length or width) than a 3X5 inch index card. This measurement allowed for the inclusion of napkins, straw wrappers, milk jugs, etc. while excluding crumbs and wrappers torn into many small

pieces. To preserve the integrity of the sample, it was determined that one student shredding a napkin into many small pieces should not be counted as several instances of littering. Before the lunch period began observers confirmed there was no litter on the lunchroom floor. After the lunch period ended, the observers walked around the lunchroom and counted and recorded the number of pieces of litter left on the floor.

Inappropriate Sitting. Inappropriate sitting included standing while eating or sitting with "back pockets off of the seat." This definition of inappropriate sitting accounted for students who stood up or walked around, turned around in seats, and hung off benches. To account for students entering and leaving the table area students were not included in the observation until after they had received their tray, placed it on the table, and sat down. Likewise, any student leaving the area was not included in the sample after his or her tray had been removed from the table. Observers counted the number of instances of inappropriate sitting by standing at the head of each table (see "A" on Figure 2) and moving between them (switching tables every 30 seconds), quickly scanning the table and counting the number of students sitting inappropriately at their table, and clicking the button on a manual hand-held counter. This procedure allowed observers to count each table once every 2.5 minutes.

Running. Running was defined as having both feet leave the floor at the same time within the observation area. This definition of running included other inappropriate behaviors such as hopping, skipping, and jumping, which would have increased the chances of students spilling food or falling and hurting themselves. Boundaries were set near both cafeteria entrances and extended for 30 feet (approximately the point where students picked up their trays) into the cafeteria. Two independent observers, one placed near each line (see "B" on Figure 2) recorded instances of running for both lines (teachers took turns bringing their classes into the lunch room, switching off between sides). Observers used a two column data collection sheet with time (in two minute increments) listed down the side. As the students entered the cafeteria observers watched them and counted instances of running that occurred. At the end of each two-minute interval, observers recorded the total number of instances of running for each side in the appropriate column. At the end of each observation session the total number of instances of running were added together to yield daily total. The orderly entry of the students into the lunchroom allowed observers to collect data from two separate vantage points.

Results

Daily totals of inappropriate behavior for each condition were plotted and visually analyzed for within and between phase differences. Figure 2 shows the differences in the data over the course of the intervention. Litter in the lunchroom decreased from an average of 34.3 pieces during baseline to an average of 1.3 pieces during the intervention phase, representing a total decrease of 96%. Inappropriate sitting in the lunchroom decreased from an average of 65.5 during baseline to 23.3 during the intervention phase, representing a total decrease of 65%. Lastly, running in the lunchroom decreased from an average of 34 instances during baseline to 8.5 instances during the intervention phase, representing a total decrease of 75%.

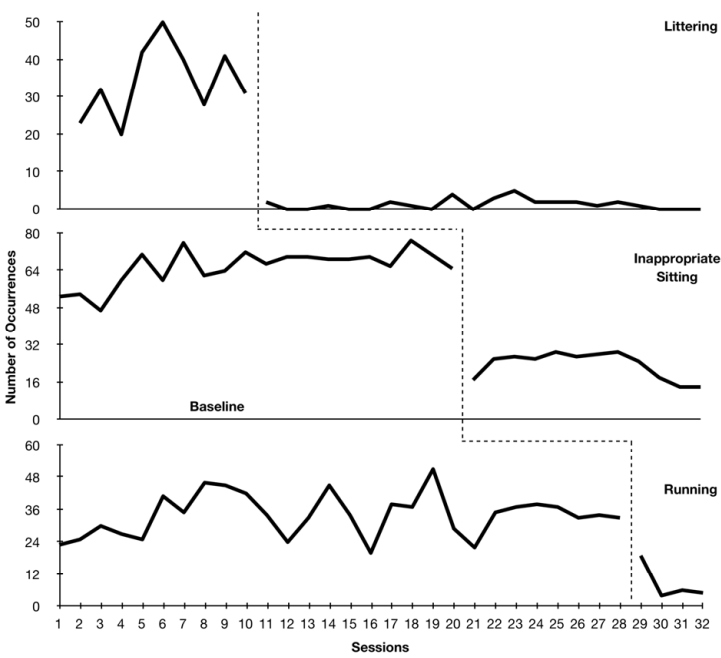


Figure 2. Multiple-baseline design across three target behaviors (littering, inappropriate sitting, and running).
Note: Vertical lines distinguish between baseline conditions and experimental conditions.

Interobserver agreement

In all conditions a second observer, who followed the same protocol as the primary observer, conducted interobserver agreement checks. During interobserver agreement checks, two independent observers recorded each target behavior. In the littering condition, each observer made sure there was no litter before the lunch period began. After the lunch period ended, each observer independently counted and recorded the number of pieces of litter left on the floor. Interobserver agreement checks were taken across 43% of the total number of data collection sessions and averaged 98% agreement. In the inappropriate sitting condition the two observers stood at opposite ends of each table and used the standard data collection procedure as previously outlined. Interobserver agreement checks were taken across 28% of the data collection sessions and averaged 90% agreement. In the running condition the additional observer was stationed on top of the bleachers on the south end of the lunchroom where it was possible to get a good view of both lines of students entering the lunchroom. Interobserver agreement checks were conducted on 28% of the trials and averaged 87% agreement. In all conditions, percent agreement was calculated by dividing the two frequency totals, one collected from each observer, and multiplying by 100 (smaller/larger $\times$ 100).

Procedural integrity data were taken on 61% of the observation days over the course of the intervention. These data were collected by the primary lunchroom observer, who randomly chose two of the lunchroom staff members and checked that 1) a Praise Note was delivered at least one time during a three-minute period, 2) the praise procedure was followed as outlined in the training, and 3) the Praise Note was filled out correctly. The staff members met expectations on 100% of the observations, and there were no instances where it was necessary to retrain the delivery procedures.

Discussion and Conclusions

Study Conclusions

The data presented here demonstrate that the Praise Note system presented in this study provides a successful method for decreasing inappropriate behaviors in an elementary school lunchroom. The authors believe the program's success can be attributed to the incorporation of several evidence-based strategies into the development and implementation of the Praise Note system. These strategies include establishing a common language to communicate behavioral expectations through direct teaching and posted signs in the lunchroom. Second, the basis for the program included the delivery of praise, specific

verbal commendations, as a social reinforcer. Third, participating school personnel were required to use active supervision by consistently moving around the lunchroom and interacting with students. Fourth, the Praise Notes delivered in this system represented token reinforcers that could be delivered immediately and then exchanged for other reinforcers later. Fifth, the Praise Note system utilized the principles of DRI by selectively reinforcing positive behaviors that were incompatible with the negative target behaviors. This collection of evidence-based strategies, implemented as a complete program, was critical to achieving the outcomes documented previously.

Limitations

In utilizing these results to improve current educational practice it is important to consider the limitations in the study's design and data collection. One limitation of this research is the inability to isolate the impact of each component of the Praise Note system on the target behaviors. This limitation precludes any recommendation regarding which strategies could be separated and used as a stand-alone intervention, although many of the cited studies demonstrate the effectiveness of these strategies when implemented in other settings and with unique dependent variables. Since it was a goal of the research to develop a simple and sustainable behavior support program, future research studies could look at this limitation as an opportunity to further refine and simplify common area behavior support interventions through a systematic investigation of which components of a Praise Note system yield the greatest impact on target behaviors.

It is also important to note that this study does not provide evidence of the intervention's impact on appropriate student behaviors. It can be inferred that appropriate student behaviors (i.e. sitting quietly, walking in the lunchroom, and not littering) increased as a result of the intervention but data were not collected to directly substantiate this claim. Although evaluating these results of the Praise Note system was not in the scope of this research project it should be investigated in future research studies. Anecdotal evidence, in the form of a short survey, collected from teachers and lunchroom staff suggests that improved student-adult relationships were a positive side effect.

The authors of this research suspect that the unique delivery method of the Praise Notes may have contributed to these positive outcomes. When delivering a Praise Note, school faculty and staff members provided students with individualized, specific praise in addition to the tangible Praise Notes. This delivery method ensured that each student who received a Praise Note, even those whose names were not selected in the drawing, received some kind of recognition

for their appropriate behavior and had a positive interaction with school personnel. Although those specific results were not documented in this study, the principal, lunchroom monitors, custodian, and regular lunchroom staff regularly made positive comments about specific students whose lunchroom behavior had improved. They also made general statements about the improved overall attitude in the lunchroom and students' willingness to encourage peers to engage in appropriate behaviors. The impact of increased opportunities to engage in positive interactions with peers and adults is a critical area outcome that warrants future investigation.

Implications for Practitioners

The current research provides an examination of the substantial impact a Praise Note system, designed and implemented as outlined here, can have on student behavior in school common areas. In addition, the authors' believe that practitioners can glean important themes from this research regarding the intensity of praise note systems, the types of reinforcers that can be used, and the impact a Praise Note system can have on teacher-student relationships.

Praise Note, or other ticket-based, systems are used by many schools as part of an effort to improve student behavior, but the available literature contains limited guidance on the intensity necessary for these systems to produce noticeable change. Because the number of Praise Notes delivered in this study was not systematically varied, we cannot say what the optimal intensity for effective systems should be. We do know, however, that over the course of the study and throughout the rest of the school year the number of Praise Notes delivered in the lunchroom averaged 110 per week, meaning that approximately 55% of students received a Praise Note each week. Lunchroom staff members reported that this number did not seem unreasonable, especially since they were eager to have a method for rewarding students for appropriate behaviors. It is also likely that the teacher reinforcement component was a factor in maintaining a high, steady rate of Praise Note delivery. The intensity of Praise Note delivery was sufficient to show definite change in all three of the target behaviors. It is possible that less intense interventions may show similar results, but further research on the topic is necessary before any recommendations to reduce intensity can be made.

Evidence from this study suggests that programs may enjoy increased success if multiple types of reinforcers are used. The reinforcers used in this study included social reinforcers (teachers praise and office staff recognition), token reinforcers (tickets), public posting (posting of Praise Notes on the board in the main hallway), group

rewards (parties and extra recess), and tangibles (erasers, pencils, folders, etc.). This program presents a method for schools to provide a variety of reinforcers in a practical manner.

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Designing and Implementing Group Contingencies in the Classroom

A Teacher's Guide

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The second-grade team was stumped. At the beginning of the year, team members had joked about their angelic students, but the honeymoon was over. There were not too many big behavior problems, but, as Mr. Finch put it, there was enough going on to eat into instructional time. Students called out, talked during seatwork, and, most irritating, constantly yelled out the teacher's name to get attention. When Ms. Smith brought up student behavior problems at the team meeting, there was a sigh of relief; at least everyone was having the same problems. Ms. Smith said, "It feels like I'm constantly putting out fires instead of teaching!" The rest of the team agreed.

The team decided to invite the special education teacher, Ms. Strong, to their next meeting. They explained their problems and nearly begged Ms. Strong for help. Ms. Strong was more than happy to help; most teachers asked for her guidance about individualizing instruction, but she was also well versed in classroom management. "One thing we're worried about," said Mr. Finch, "is that we all are having behavior problems at different times. I need help during large group instruction but Ms. Smith wants help when she's working with small groups of students."

After listening to the team, Ms. Strong had an idea. "Have you heard of a group contingency?" she asked. She looked out at the blank faces. Ms. Strong laughed. "You'll love this. It's a behavior management tool that is completely adaptable to your needs, and research shows that it works!"

A group contingency is a classroom management system designed to proactively support appropriate classroom behavior. Typically, groups or teams of students are rewarded for exhibiting appropriate or desirable classroom behaviors rather than being punished or reprimanded for exhibiting inappropriate or undesirable behaviors. Research on behavior management strategies that incorporate group contingencies have found that these programs are generally effective for improving student academic and social

outcomes and that they may change teacher behavior, increasing teachers' attention to, and praise of, appropriate classroom behaviors (Barrish, Saunders, & Wolf, 1969; Kamps et al., 2011; Wills et al., 2009; Wills, Iwaszuk, Kamps, & Shumate, 2014). Group contingencies are a peer-oriented program that focuses on preventing problem behavior by reinforcing appropriate behavior. Better yet, they are efficient, easy to use, and adaptable to match different classroom contexts. Broadly, these preventative interventions include three primary components: establishing classroom expectations, explicitly teaching the expectations, and reinforcing the established and taught expectations. Group contingencies are particularly useful as more students with disabilities are included in general education settings and as special educators work collaboratively with general educators to support student behavior within tiered support systems.

Students With Disabilities in the General Education Classroom

Sixty percent of students with disabilities spend more than 80% of their time in general education classrooms (National Center for Education Statistics, 2013), and students with disabilities may exhibit more challenging behavior than their peers without disabilities (Blacher, Howell, Lauderdale-Littin, Reed, & Laugeson, 2014). However, many teachers may not be prepared in classroom management practices (Greenberg, Putman, & Walsh, 2014). As special education teachers work with general education teachers to support students with disabilities in general education classrooms, special education teachers likely need to address student behavioral needs in addition to academic needs. Group contingencies are easy to implement in the general education classroom to support the behavior of students with disabilities, and they offer an opportunity for collaboration between special and general education teachers.

Current Trends: Multitiered Systems of Support

In recent years, school districts have increasingly implemented multitiered systems of support (e.g., response to intervention, positive behavior supports) in an effort to meet the academic and social needs of all students. School-wide Positive Behavioral Interventions and Supports (SWPBIS) is a decision-making framework that guides the selection, integration, and implementation of evidence-based academic and behavioral practices for improving academic outcomes for students (National Center of Positive Behavioral Interventions and Supports, 2013). The SWPBIS framework promotes a three-tiered approach with Tier 1 supports that are universal and available to all students (Sugai & Horner, 2009). Within the SWPBIS framework, group contingencies are designed for implementation at the Tier 1 level but allow for adaptations to be made that are specific to individual student needs. For example, if a student with a behavioral disorder or attention-deficit hyperactivity disorder is in a classroom with a group contingency in place, the behavioral goals on the student's individualized education program (IEP) can be incorporated into the universal program. Effective Tier 1 programs may help to prevent behavior problems frequently exhibited by students with disabilities and help these students be more successful in general education classrooms. Group contingencies are designed to be pragmatic and easy to implement; they support all students in the classroom; and they provide an opportunity for special education teachers to work collaboratively with general education teachers. The adaptability of universal behavior programs allows for student-specific modifications to be made within the context of the classwide intervention.

Although group contingencies are receiving more attention as Tier 1 interventions under SWPBIS, group contingencies are not a new development. In fact, a number of standard protocols exist, including the Good Behavior Game (GBG; Barrish

et al., 1969) and the Class-wide Function-Based Intervention Teams (CW-FIT; Kamps et al., 2011; Wills et al., 2009). As part of GBG, the teacher divides the students into teams and defines acceptable and unacceptable classroom behaviors. Each time that a member of the team exhibits an undesirable behavior, the team receives a mark. After receiving a specified number of marks, team members do not receive access to a reward. In this way, the group contingency supports the relation between GBG and improved student behavior.

4. Decide how you will award points.
5. Choose who awards the points.
6. Determine a schedule.
7. Select rewards.

Choose the Target Behaviors

The first step in designing a group contingency is choosing the behaviors that you want students to exhibit. The target behaviors should be clear, positively stated, and easy for students to understand. You may already have these behaviors posted as rules in your classroom, but make sure that they

‘raise your hand to get the teacher’s attention’ instead of ‘don’t call out’?

“Sounds great,” said Mr. Finch. “I think that my students could use a refresher on the rules.”

“That’s a really good point,” said Ms. Smith. “I know that all of you went over the rules at the beginning of the year, but I think that some students may benefit from explicit instruction on the rules. Instead of just reading them, try showing the students what following the rules look like. Provide some examples and nonexamples.”

The other teachers agreed to go over the rules with more modeling and post the rules on their classroom walls.

Broadly, these preventative interventions include three primary components: establishing classroom expectations, explicitly teaching the expectations, and reinforcing the established and taught expectations.

Another “off the shelf” group contingency is CW-FIT. The teacher provides explicit instruction to students on desired behaviors before dividing them into teams. The teacher provides verbal praise and points to teams exhibiting the desired behaviors at prechosen intervals. At the end of the game session, the teams that have met the predetermined point goal access a small reward. CW-FIT is emerging as an effective group contingency for improving student on-task behavior (Kamps et al., 2011; Wills et al., 2014).

Designing a Group Contingency

Some teachers may prefer to create their own group contingency programs instead of relying on standard protocols, such as GBG and CW-FIT. Although the existing programs are adaptable, teachers can easily design group contingencies to match the needs of their classrooms and students. In this section, we walk through seven steps to design a group contingency for your classroom:

1. Choose the target behaviors.
2. Choose your groups.
3. Determine how groups earn points.

focus on the behavior that you want students to exhibit. Take some time to explicitly teach or reteach these behaviors to students, and make sure that the behavioral expectations are posted in your classroom in a spot that is easy for students to see and for you to refer to. Include modeling of the desired behaviors and practicing the behaviors as a group. Students who have behavioral goals in their IEPs may benefit from having these goals incorporated into a group contingency. If you are providing support for students’ behavior in a general education classroom, you should include a specific rule that addresses the needs of the student with a disability whom you are supporting. This way, the student is not singled out but the individual’s behavioral needs are still addressed.

The teams had developed four rules at the beginning of the school year: use kind words, keep your hands and feet to yourself, don’t call out, and do your work. “These are good rules,” said Ms. Strong, “but the directions for a group contingency say it’s important to have the rules positively stated. How about

Choose Your Groups

Group contingencies rely on teams of students to support behavior. Not only does this approach promote cooperation, but it also prevents the singling out of students for undesirable behavior and allows you to address the behavior of multiple students simultaneously (Gresham & Gresham, 1982; Little, Akin-Little, & O’Neill, 2015). Group students into similarly sized teams based on proximity. You may want to create the groups according to classroom seating arrangements or instructional groups. However, if instructional grouping results in a team that includes all the students with challenging behavior, you may need to reconsider using these teams. Part of the success of group contingencies is that peers act as models of appropriate behavior and can provide prompts to students within the group. If all students in the group struggle with behavior, they may not be able to support one another. In addition to the composition of student groups, it is important that students clearly see the team to which they are assigned, and it is easier for the teacher if the teams are clearly defined.

Sometimes, you may be working with only a few students at a time—for example, during pullout (resource) instruction. You could elect to use the entire class or pullout group as one group, or you can set up a behavior management system using these same steps but with

teams consisting of only one student. The students will still have the same point goal and reward. Sometimes this is most appropriate depending on the setup of your classroom and the characteristics of your students.

After target behaviors are selected, teams needed to be created. “This part is easy!” said Ms. Smith. “I want to use a group contingency during small group instruction, so I will use my instructional groups.” Mr. Finch’s

The first step in designing a group contingency is choosing the behaviors that you want students to exhibit. The target behaviors should be clear, positively stated, and easy for students to understand.

situation was a little more difficult. He planned to use a group contingency during the morning meeting, when all the students were sitting on the carpet. He had one student, Sarah, who seemed to enjoy getting into trouble with the other students. He expressed his concerns to Ms. Strong, who had a number of suggestions. “It can be tricky to have the students in teams during carpet time. What if you have them sit in rows and make each row a group? I agree that having Sarah on a group might be a problem. What about using an independent group contingency? You can have goals for the whole class but meeting the goal depends on the individual student. Maybe try that out with Sarah?”

Mr. Finch looked worried. “I’m not sure I can track the progress of all 17 students at once while teaching.”

“Good point,” said Ms. Strong. “I don’t love singling students out, but maybe for now Sarah is on a team of one? That might make it more manageable to award points. She’ll still have the same goal and follow the same rules, but this way you can give her more individualized attention when she’s meeting the expectations. See how Sarah does, and maybe she can join a group later.”

Determine How Groups Earn Points

Studies of group contingencies include three ways that groups can earn points: interdependently, dependently, or independently. An *interdependent group contingency* is when everyone in a group must be exhibiting the desired behavior to earn a point. If one student in a group of four is not following the rules but the other three students are, the team would not receive a point.

In another approach, a group earns points as part of a *dependent group contingency*, where certain members of the team represent the entire group. For example, you may be concerned with the behavior of only a specific student. When this student is following the rules, the entire team earns a point without depending on the behavior of the other teammates. Finally, an *independent group contingency* is when all students follow the same rules, have the same goal, and access the same reward, but they obtain the reward based on their individual behavior (Theodore, Bray, & Kehle, 2004). Independent group contingencies are particularly useful if some students sabotage the progress of a group. In this type of contingency, students are rewarded according to individual behavior but with the same expectations for all students. Of course, tracking points for an independent group contingency with a large class may be burdensome, but this approach is useful during small group or pullout instruction. These three types of group contingencies may improve student behavior (Ling, Hawkins, & Weber, 2011; Theodore et al., 2004; Wright & McCurdy, 2012); however, interdependent group contingencies are usually preferable. Interdependent group

contingencies make clear that behavioral expectations are for all students while encouraging teamwork and are easier for teachers with a large number of students to successfully implement.

Decide How You Will Award Points

Now it was time for the teachers to decide how to assign points. The team decided to use an interdependent system where everyone on the team has to exhibit the behaviors for the team to earn a point. “This makes the most sense to me,” said Mr. Finch. “It should help the students work together and support the behavior of other team members.”

“How many points should the goal be?” asked Ms. Smith. “I want it to be challenging but obtainable.”

“Excellent point,” said Ms. Strong. “At the beginning, I would start with an easier goal. This will let the students learn the gain and access the reward. We don’t want them to feel like the goal is impossible and experience failure at the beginning.”

“You know, it might be good if we use the data from Sarah’s functional behavior assessment to help figure out the goal,” Mr. Finch suggested to Ms. Strong. “That way I can look at her current level of disruptive behavior and set a class goal that isn’t too hard for Sarah to meet.”

The team agreed that an easier goal could help students start on the right path and that the goal could become increasingly difficult over time.

You can award points using a reward system or using response cost. A reward system is when teams earn points when they are exhibiting the desired behaviors. This system provides recognition to students for following rules and is a positive approach to classroom management. Conversely, teams lose points for each undesirable behavior under a response cost system. Response cost can be useful when students are engaging in more severe problem behavior, but generally a reward system is a better and more positive approach. The response cost approach provides attention and reinforcement

(depending on the student) to the behavior that you are trying to eliminate. Attention frequently acts as reinforcement and can maintain undesirable student behavior (Kodak, Northup, & Kelley, 2007). In contrast, a reward system provides attention and reinforcement to the behaviors that you want students to exhibit.

"It seems to me that rewarding points is a lot more positive than response cost," said Ms. Smith. "I agree—and it better fits the school approach to positive classroom management," said Ms. Strong. "What about when students break the rules? Can I take away points?" asked Mr. Finch.

"I wouldn't," said Ms. Strong. "It sends a mixed message to students, and it gives attention to the behavior that you don't want to see." The group decided to reward points to teams for following the rules and agreed not to take away points.

Choose Who Awards the Points

The next step in designing a group contingency is deciding who keeps track of team behavior and awards points to each group. Groups may self-monitor (Briesch & Chafouleas, 2009), or the teacher may monitor and reward points. Under self-monitoring, when it is time to award a point, the group reflects on if it was following the rules and records if it earned the point on a group point chart. Self-monitoring is a good approach when you trust that students can successfully manage their behavior and when the students respond well to the other members of the team.

If you feel that your students cannot be trusted to reliably observe and assign points, teacher monitoring is a better approach. Under a teacher-monitored approach, the teacher decides if each team has earned the point. Teachers often rely on teacher monitoring when implementing a group contingency program and transition into a self-monitoring approach once students understand the program and the behaviors needed to earn points. Teachers can

also use classroom support personnel (e.g., paraprofessionals, parent volunteers, student teachers) as another strategy for keeping track of awarding points.

When deciding if points should be teacher or student monitored, Mr. Finch and Ms. Smith said in unison, "Teacher monitored." Ms. Strong laughed. "That should work, but you may want to let students self-monitor once they get a hang of the game. I've seen it work!"

"We may need to slightly adapt this for Jonathan," Ms. Strong said to Ms. Smith. Jonathan has a visual impairment, and Ms. Strong was concerned that he would not be able to see the class point chart. "Why don't I make him a portable point chart so that he will be able to track and see his group's points?"

"Oh good idea!" said Ms. Smith. "Actually, it might be helpful if you come in the first day that I use a group contingency and help with his group. Could you show Jonathan's group members how to mark the team points on his personal point chart? I definitely want him to be able to see his progress toward the goal."

Determine a Schedule

There are three components to this step in designing a group contingency: length of the contingency, frequency of awarding points, and when your students access their reward. First, think about how long the contingency should last. You may want the group contingency to take place during a particularly challenging transition, small group instruction, or a specific time of the day. The program works best during a time with a specified start and end, rather than all day. When first implementing a group contingency, you will want to plan for a shorter amount of time so that students more quickly access the reward. Over time, you can extend the length of the group contingency.

Second, consider how frequently you want to award the teams points. There may be natural times during a lesson to stop and award points (e.g., when

students move to new stations), or you may want to use a timer to specify a interval of time. It is best to use a shorter amount of time (i.e., to provide more opportunities for teams to earn points) when you are beginning implementation. This provides students with more opportunities for behavioral feedback and experiencing success. The length of time between awarding points can vary according to student need and what fits best with your lesson.

Third, consider when the students will access the reward. Immediate access is preferred over waiting until later in the day or week to access the reward. When deciding when to play the game, keep in mind a good time for students to receive the reward. If you do need to provide the reward later—for example, when providing additional recess or lunch in the classroom—make sure to clearly state which teams won the reward and when they will receive the reward.

Ms. Smith worked through her schedule first. She planned to use a group contingency during small group instruction. The groups rotated every 15 minutes, but she was worried that this was too long of an interval for second-grade students. She decided to set a timer for 5 minutes and give out points every time the timer went off during the entire 45-minute period of small group instruction. Small group instruction was right before lunch, so she planned to give 3 minutes between instruction and lunch to give the winning teams their rewards.

Mr. Finch planned to use a group contingency during the 30-minute morning meeting. He felt like some of his students would have a hard time following the rules at first so he decided to set a timer for every three minutes, hoping to make the interval longer as students adjusted to the game. The end of morning meeting also seemed like a good time to give students their rewards.

Select the Rewards

The rewards for a group contingency do not have to be large or expensive.

Remember: All teams can “win,” so you will potentially be providing a large number of rewards. To identify rewards, use a preference assessment to determine what students like. Preference assessments ask students to identify activities and items that they like (Northup, 2000). Preference assessments can be administered individually or as a group in either oral or written form. Some examples of creative rewards include watching a short YouTube video, having a 3-minute dance party, doing classwork in a colorful pen, and taking shoes off during the next activity. It is important to vary the reward to keep a group contingency exciting and interesting to your students.

The results of the preference assessment may show that some students have very specific interests and may need to be working for a reward different from that of the group. This is a helpful adaptation and may be particularly important for students with disabilities. For example, the second author taught a student with autism who was not interested in a noisy class dance party but loved reading about elevators. This student had a separate reward from the class; instead of a dance party, he used the time to read about elevators on Wikipedia. The student still participated in the game but had a personalized reward that was more reinforcing to him than the group reward.

Begin the program for the day by stating the reward that students will be working toward. We suggest two approaches to choosing the reward. First, you can choose the reward. This prevents any discussion among students over what is fair or what the reward should be for the day. Second, some teachers write all the possible rewards on slips of paper or popsicle sticks and choose a random reward. This approach can add another fun element to a group contingency.

Another adaption to stating the reward at the beginning of a group contingency is to have your students work toward a surprise reward. Sometimes a group of students will not find a reward as motivating as other students. The surprise reward can add a

fun element of suspense to the game and prevent students who are not interested in the selected reward from failing to buy in to the game and practice exhibiting desirable classroom behavior.

The team brainstormed possible rewards. “It’s important to me that they’re free and don’t take a lot of time,” said Ms. Smith. The team came up with a number of activities and created a survey for students to circle the activities that they liked the most. Ms. Strong volunteered to print the students’ chosen activities on cardstock so that the teachers could cut the rewards out and select them from a bag

can make a group contingency more effective for these students.

If you find that your students seem disinterested, revisit your rewards, and make sure that the reward is desirable for your nonresponding student. Also revisit the goal and the frequency with which points are awarded. Setting an easier, more obtainable goal can provide an opportunity for students to experience success. Awarding points more frequently presents more opportunities for reinforcing the desired behavior and provides more frequent reminders to students to correct their behavior. If a team consistently fails to meet the goal, try

Instead of using a reprimand, praise a student nearby the misbehaving group, or use a bonus point to remind groups of what they should be doing.

each day. Mr. Finch had an important question for Ms. Strong: “What should I do with the teams that don’t win?”

“Teachers ask me this a lot,” said Ms. Strong. “I’ve noticed that students can be upset about this when they are first learning the game. Make sure to let them know that they can earn the reward tomorrow by following the rules. Don’t dwell on it too much, and make sure that the teams that don’t win have something to work on while the others receive a reward. I think you’re ready to implement your group contingency!”

Keys to Success and Potential Pitfalls

Teachers may encounter a few common challenges when designing and implementing a group contingency. First, some students may not seem to respond. Consistent with the SWPBIS framework, a small number of students may need more intensive and individualized interventions, such as check-in/check-out (e.g., Todd, Campbell, Meyer, & Horner, 2008) or even individualized behavior support plans, but there are adaptations that

using bonus points in between normal point intervals to draw attention to desirable behavior.

Although implementing a group contingency can allow for more opportunities to provide attention to desired behaviors, implementation may require some adjustments to teaching. For example, it can be difficult to ignore students who are not following the rules, while providing positive feedback to others; it is much easier to focus on undesirable behavior because it can be very disruptive and distracting. If you find that students’ behavior has not changed as a result of a group contingency implementation, pay attention to the number of reprimands that you are using. Instead of using a reprimand, praise a student near the misbehaving group, or use a bonus point to remind groups of what they should be doing. In doing this, you are highlighting—in front of the whole class, including the misbehaving student—the behaviors that you want to see and demonstrating how rewarding it can be to exhibit desired classroom behavior. When teachers first start their group contingencies, we recommend they be

cognizant of the importance of changing their own teaching behaviors (e.g., praising students who are doing what you want them to do, instead of scolding students who are off task). This may be even more important when you are helping others (e.g., general education teachers, other special education teachers). However, increasing attention to positive behavior and allowing opportunities for success by setting achievable goals are keys to a successful group contingency.

The second-grade team was meeting a few months later when Ms. Strong walked by. They called her into their meeting. "I can't thank you enough!" said Ms. Smith. "The first couple days of a group contingency were a little rough, but now my class loves it, and I finally have enough time to teach."

"And I no longer have students constantly calling out my name," shared Mr. Finch. "I like how flexible the group contingency is. I've even started playing it as a surprise during other lessons when the students need a little extra behavior support."

Conclusion

Group contingencies are a positive, proactive classroom management technique that works well as Tier 1 of a multitiered system of behavior support. These programs are adaptable to student and classroom needs and work well to support the behavior of students with disabilities in general education classrooms. Off-the-shelf programs exist, but teachers can also follow these seven steps for designing a group contingency to match their needs as well as the needs of their individual students.

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Shaping With Successive Approximations

Differential reinforcement, as previously discussed, is a technique for increasing the strength of selected responses that are members of a response class. To develop a response that differs significantly from members of an existing response class, we can use the procedure of **shaping with successive approximations**. The shaping procedure involves the use of differential reinforcement to strengthen members of one response class. After these responses are performed consistently, the criterion for reinforcement is shifted to another response class that more closely approximates the target response to be developed. This procedure is repeated until responses are performed and reinforced in a response class that includes the target response.

Shaping with successive approximations is a procedure used to develop a new behavior or one that rarely occurs. A shaping procedure was used as one part of a treatment program to reinstate speech in Justin, a 10-year-old boy who stopped speaking after witnessing his mother’s death in a car accident. At first, Justin was reinforced with candy and praise for any speech sound he made. After he consistently made speech sounds, the criterion for reinforcement was shifted to speaking words. After Justin consistently spoke words, the criterion for reinforcement was shifted from words to phrases. In this way, Justin’s speech was shaped using successive approximations until he spoke in complete sentences.

In shaping with successive approximations, a series of initial and intermediate behaviors are established in successive approximations to the desired target behavior. The initial response that is reinforced bears some resemblance to the target behavior (e.g., speech sounds and speaking in sentences) so that the intermediate responses can be progressively shaped toward the target behavior. To shape a behavior, the practitioner takes the following steps:

1. Specify the target response (desired behavior).
2. Specify the positive reinforcer(s) to be used.
3. Specify initial and intermediate responses.
4. Reinforce the initial response each time it occurs and withhold reinforcement from other responses until the initial response is performed consistently.
5. Shift the criterion for reinforcement from the initial response to an intermediate response.
6. Reinforce the intermediate response until it is performed consistently, then shift the criterion for reinforcement gradually to other intermediate responses that are increasingly similar to the target response.
7. Reinforce the target response when it is performed.

Justin’s therapist used shaping with successive approximations to reinstate Justin’s speech as follows:

1. The target behavior was for Justin to speak in sentences in response to questions the therapist asked.
2. The positive reinforcers given to Justin were candy and verbal praise from the therapist.
3. The initial response criterion included any speech sound. Intermediate responses included words and phrases that Justin spoke in response to questions the therapist asked.
4. Initially, any speech sound Justin made was reinforced.
5. When Justin made speech sounds consistently, the criterion for reinforcement was shifted to words. Justin was required to speak words before receiving the candy and praise.
6. When Justin was speaking words consistently, the criterion for reinforcement was shifted to another intermediate response class, phrases, which was the next approximation to the target behavior. Speaking in phrases was reinforced until Justin spoke in phrases. The criterion for reinforcement was then shifted to speaking in sentences.
7. Justin was reinforced for speaking in sentences in response to questions the therapist asked. At that point, Justin’s speaking was reinforced by many people in his environment and further treatment could proceed.

A therapist employed shaping with successive approximations to teach Barbara, a teenager with muscular dystrophy, to walk with crutches instead of using a wheelchair. Although Barbara could be more independent and active on crutches, she initially refused to use them, preferring to rely on her mother or sisters to push her wheelchair. The therapist used the following steps to shape Barbara's use of crutches:

1. The target response was walking 50 steps on crutches.
2. The positive reinforcer was praise (e.g., "good," "very good," and "that's it!").
3. The initial response was movement toward the crutches, which were placed within Barbara's reach. Intermediate responses included touching the crutches with her hand, holding the crutches in her hand, using the crutches to raise herself from the wheelchair, standing up with the crutches properly positioned, and taking from 1 to 49 steps on the crutches.
4. Initially, when Barbara made any movement toward the crutches, she was reinforced with praise.
5. When Barbara reached toward the crutches each time they were placed within her reach, the criterion for reinforcement was shifted to the next intermediate response, touching the crutches.
6. Touching the crutches was reinforced until the touching responses were performed consistently. This procedure of reinforcing one response until it was performed consistently, and then shifting the criterion for reinforcement to the next intermediate response, continued until the target behavior, walking on the crutches, was performed and reinforced.
7. Barbara was reinforced for walking 50 steps on the crutches.

Shaping with successive approximations involves a gradual process in which a response must be developed at one level before reinforcement is shifted to the next level of approximation. After a desired response is performed, it is reinforced immediately to ensure that reinforcement is given only for appropriate responses. If the criterion for reinforcement is shifted too quickly to the next level, or if insufficient reinforcement is given, the response could extinguish. If a response receives too much reinforcement, however, it can become fixated at that level so that it is difficult to develop the next intermediate response. The shaping procedure relies on reinforcing responses that the individual is currently performing and gradually shifting the criterion for reinforcement to intermediate responses until the target response is performed.

Instructions and prompting are often used with shaping to facilitate the acquisition of new behaviors. Although shaping alone is useful with individuals who do not follow instructions, the shaping procedure becomes more effective when instructions or prompts are given at each step (see Chapter 6). Physical guidance can be used, where appropriate, to facilitate shaping. Demonstrations of the target behavior by a model can also be used to promote rapid development of desired behaviors (see Chapter 8).

Shaping with successive approximations can also include the use of punishers (see Chapter 9), as occurs in the children's game of "hot and cold." In this game, an object is hidden from a child who has left the room. When the child returns, he or she must find the object with the help of only two types of feedback from the rest of the group: When the child moves closer to the hidden object, the group says "hot" (positive reinforcer), and when the child moves away from the hidden object, the group says "cold" (punisher). The group uses "hot" to increase or strengthen responses toward the hidden object and "cold" to decrease or weaken responses away from the object. The child performs responses that result in the others saying "hot" and stops performing responses that are followed by their saying "cold." Thus "hot" serves as a positive reinforcer for performing responses that lead to the hidden object, whereas inappropriate responses are weakened by "cold."

Summary

1. Differential reinforcement involves the use of both positive reinforcement and extinction. One response is positively reinforced and increases in strength or likelihood of occurrence, whereas other responses are extinguished and decrease in strength or likelihood of occurrence.
2. Response differentiation is the result of differential reinforcement that involves selective positive reinforcement of certain responses and extinction of others. The reinforced responses become differentiated—that is, they are performed frequently to the exclusion of the extinguished responses.

3. The words *response* and *behavior* actually refer to a class of responses rather than a single response. When one member of a response class is reinforced, all responses in that class are also strengthened. Therefore, one member of a response class represents a subclass of responses rather than a single, discrete response.
4. The DRO procedure can be used to decrease undesired behaviors by reinforcing any behaviors other than the undesired one.
5. Differential reinforcement of incompatible behavior (DRI) is used when a behavior that interferes with the target behavior is specified to receive reinforcement, thereby reducing the likelihood that the undesired behavior will be performed.
6. Shaping with successive approximations is a procedure for establishing a new behavior or one that rarely occurs.
7. Shaping with successive approximations involves using differential reinforcement to strengthen members of one response class and then shifting the criterion for reinforcement to other response classes until the desired target behavior is performed. Intermediate behaviors are reinforced and developed as successive approximations to the desired behavior.
8. Instructions, prompting, modeling, and the use of punishers can also be used with differential reinforcement to develop new behaviors.

Suggested Activities

1. With one class member who has volunteered to be a subject out of the room, identify a behavior to shape in that person without using any verbal instructions. When the subject returns to the room, another student who has volunteered to be the shaper uses a toy clicker, a whistle, or hand clapping as a reinforcer to shape the subject's behavior. The rest of the class should remain silent and refrain from providing any cues to the subject. The only instruction the subject should receive is that whenever he or she hears the clicker, the whistle, or the clapping, he or she should imagine receiving a wonderful reinforcer.
How long did it take to shape the correct response? Was the correct response performed and reinforced? Repeat this exercise a few more times with different shapers and subjects. Discuss your observations regarding the factors that facilitated or hampered successful shaping and the experience of being a subject or shaper.
2. Pair up with another class member and silently select a class of verbal responses (e.g., about food, clothing, cars, travel) that you will differentially reinforce in your partner using attention and conversation. Use differential reinforcement to narrow your partner's conversation so that he or she talks only about the class of responses you have identified. When both partners have had a turn at this exercise, discuss your observations regarding what seemed to work best in achieving the desired behaviors.

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